

Robocup - Detailed Design Report

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1 Introduction

This report describes the mechanical, electrical and software development of the robot design choice for the Robocup competition. The robocup competition requires the robot design to compete in head-to-head rounds, navigating an arena with obstacles including walls, ramps, pillars, and ridges. The robot must identify and avoid plastic weights whilst collecting metal weights. Along with these requirements, a "snitch" may be collected to achieve a large number of points. The "Sucky Sucky 9000" design was chosen to be developed. The main reasons were for the design's efficient weight collection system, its circular shape allowing for full arena traversal and its buildability.

Due to the designs' unique geometry, the mechanical design ensured that all elements would fit within the circular shape constraint. The initial concept involved a 3-level system that would split the software, electrical, and mechanical systems into distinct areas for improved buildability. The mechanical systems identified onboard were, the chassis, movement, collection, weight storage and sorting. Software systems involved general navigation, weight detection, motor control and, movement. A Fault Tree Analysis (FTA) and Finite State Machines (FSM) were used to diagnose any potential problems; with practical testing conducted to finalise any design choices and to refine manufacturing tolerancing for any possible part remanufacturing.

The chassis design was refined from the existing 3-level design that used the open beam profiles. The sorting system had to be redeveloped due to space constraint limitations and general uncertainty with the system's performance. This involved the incorporation of a revolver-like sorting system, which solved the space constraint, but complicated development and buildability. The ramp system had to be changed to suit this development, with the creation of guide walls functioning as a static guide for weights entering from the harvester.

2 Progress to Date

2.1 Overview of the Current Design

The current design and all preceding decisions were made based on the belief that a circular robot shape is an effective solution, with the strict new design guideline that the shape profile could not be interrupted. The current design is a result of both that constraint and the previous criteria identified to create a competitive robot design. Thus, completing the mechanical aspects was the main focus as to give a static platform to develop the code upon. Figure 1 below shows the completed mechanical design, with the chassis, track system, harvester, and sorter now fully complete.

2.2 Chassis and Movement

The chassis was developed from iteration of the "Sucky Sucky 9000" design concept, and was structured around utilizing a 3-level layer system. It is structured into the harvester (layer 1), motor control/weight sensing circuitry (layer 2), CPU, and environmental sensing circuitry layers (layer 3), shown below in Figure 2.

Each of these layers were made from 6mm MDF board (seen above), cut to a circular diameter of 350mm (level 1) and 380mm (levels 2 and 3) using lazer cutting (as described in Appendix E 1, 2 and 3). Using MDF accelerated the development of the levels with the lazer-cutter's fast production time, and easy availability to access the lazer-cutter through most of the day (**Requirement 5.2**). Later in the development stage, labeling component placement was introduced to the layers, which decreased assembly time for development(**Requirement 4.1**).

The levels were supported by 4 Open Beam (OB) 210mm aluminum beam profiles, with 3D printed "tensioner" profiles that would be able to slide along the beams. This "tensioner" design was made to be multipurpose, connecting the top and bottom levels with 2 extra side screws for added rigidity for the levels, shown in Figure 3.

The "tensioner" profiles were adapted to support the harvester collector system, and unique profile heights to create the correct initial offset for the track tensioning system. The printing tolerance for the outline profile had to be experimented, finding that a 0.5mm offset from the aluminium profile yielded

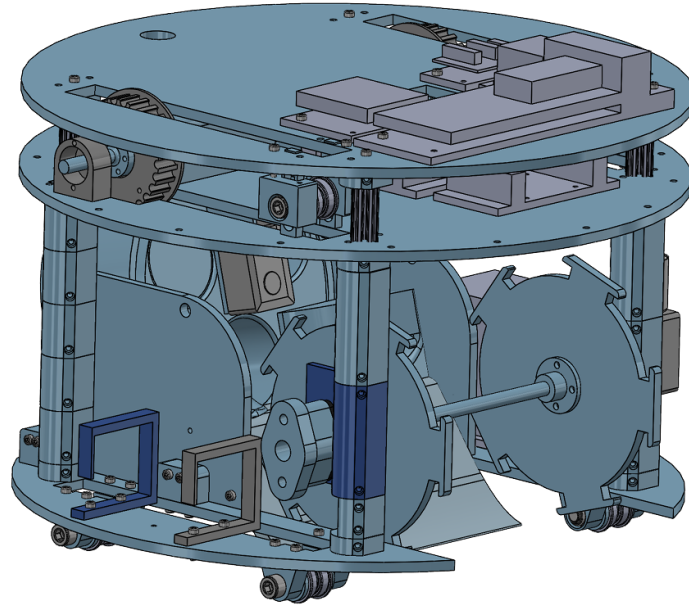


Figure 1: Overall robot mechanical design from CAD file, with included harvester, track, tensioning, and sorting systems on board

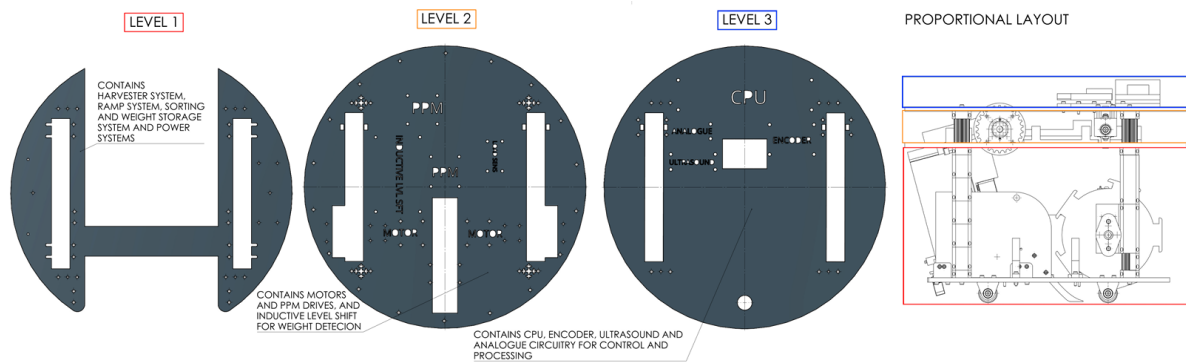


Figure 2: Level sets of the robot, with highlighted location of associated parts to those layers

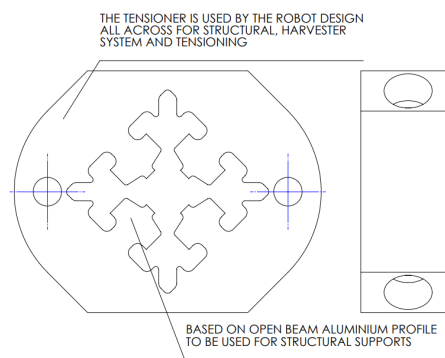


Figure 3: Tensioner profile drawing, with tolerances and dimensioning for usage across the robot (full resolution drawing is provided in the appendix)

easy slotting of the profiles and good structural rigidity (the technical drawing to this piece is provided in Appendix E 4). This rigidity was confirmed through several shake tests to work out any weak points.

For the tracks, cutouts were made into all level boards to maintain the circular design and retain appro-

appropriate track tension. Shown in Figure 4b below, the tracks were mounted onto 3D-printed "suspension", which were connected to level 1. This part created significant offset from the ground to the robot base, allowing for easy traversal over the ramp in testing (**Requirement 1.5**). The battery and power pack were moved to level 1 to improve the robots' Centre Of Mass (COM).

The meshing gear for the tracks were completely redesigned, with the intended purpose of minimizing track slip. The gears are based on the track profile to enable perfect meshing, with the centre allowing for a metal rigid flange coupling as shown in Figure 4a below. This allowed for metal-metal contact with the electric motor to reduce slippage (**Requirement 2.2**).

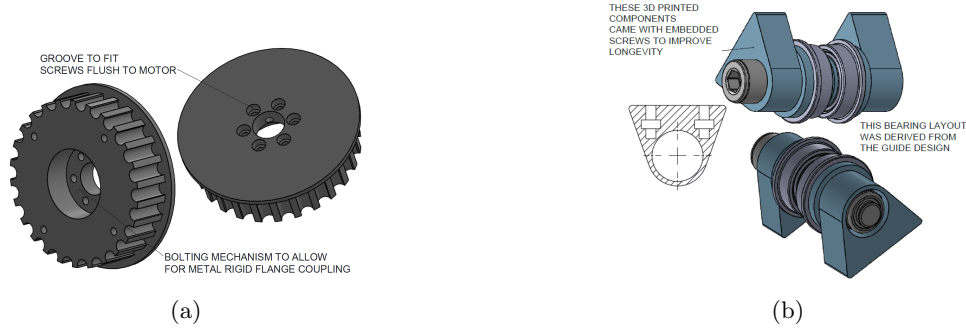


Figure 4: General details on the meshing gear (a) and suspension design for the robot's track system (b) (relevant engineering drawings are placed in Appendix A)

When undergoing movement testing, bending of the meshing gear and the distortion of the track shape was noticed when the tracks were highly tensioned. This distortion could be another source of slip, hence a support part was made to fully secure the motor component, shown in Figure 5a below. The PLA printing was done to optimize rigidity to reduce any noticeable displacement in the support beam.

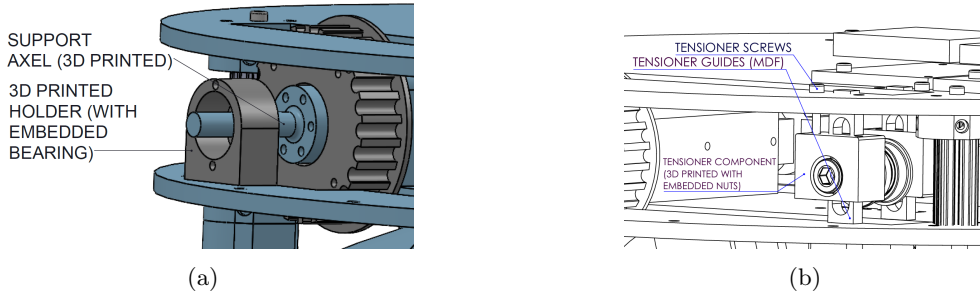


Figure 5: General details on motor axle support system (a) and track tensioner system (b) (relevant engineering drawings are placed in Appendix A)

Finally to bring the design together, the tracks needed easily modifiable tension to mechanically tune the robot as well as to make disassembly easy (**Requirement 4.1**). This system went through several iterations, settling on a 4 screw tensioner shown in Figure 5b. Guided by lazer-cut MDF guides, this special suspender would move up through screws, which stretched the track.

2.3 Collection Mechanism

The harvester system was derived from previous implementations from other robot designs, such as the 2024 winner, however with using thicker rubber bands and new motor implementation. The harvester was built around modified tensioner profiles, with a 50mm length and features to hold the KFL M8 bearing and 50 rpm encoder motor. These features were then covered by similarly shaped profiles to also perform the purpose of track tensioning. The overall result is shown in Figure 6a and Figure 6b below. The harvester wheel had considerable testing done upon it to allow for enough space for the weights to fit around the bands, but also to limiting deflection in the rubber bands to ensure constant force application up on the ramp (the final design can be found in Appendix E 5). This resulted in a 5-band system, which utilised thicker bands to allow for more potential elastic force and area of application.

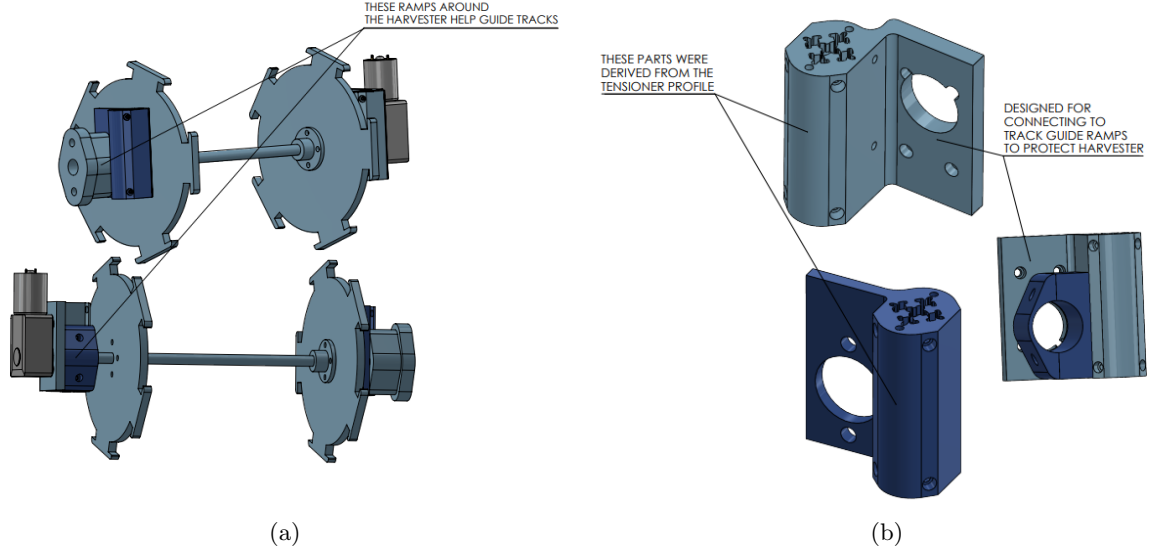


Figure 6: General details on -storage system (a) and general logic tree for programming the storage mechanism (b)

The ramp design that was previously conceived was further refined, where the ramp was designed to be partially concentric with the harvester wheel. It was found through testing that a smoother entry into the ramp, a concentric guide for the weight to climb up and the linear exit out of the ramp was the most effective design, as shown in Figure 7 below. The ramp then was combined with the sorter system, as previous ideas of a long ramp sorter was deemed unrealistic because of the circular constraint. The harvester and ramp system allowed for all weights be picked up, no matter the orientation, within the 5-second constraint (**Requirement 1.3**).

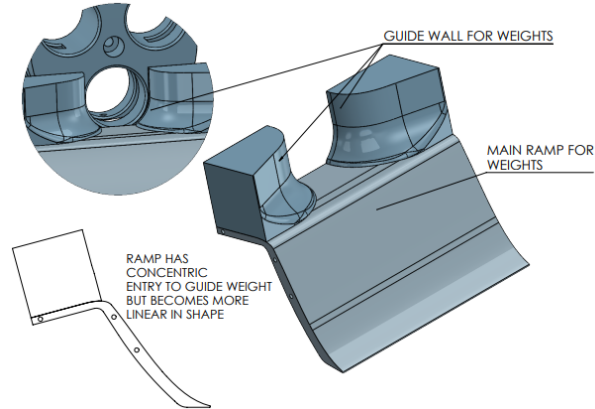


Figure 7: General details on the ramp design and weight guiding walls used

2.4 Weight Storage and sorting mechanism

The current weight storage design has required a significant redesign from the initial concept. Instead of a gravity-based pin ball sorter mechanism, a revolver cylinder-like storage system was devised, with guiding walls combined with the ramp system, shown in Figure 8a below. The weight revolver relies on a 10 RPM motor to drive the weight holder drum, with an exterior drum housing an inductive sensor and limit switches. The programming logic for the sorting system is described below in 8b. The sorter's inductive sensor allowed for efficient sorting between plastic and metal weights in 2 seconds from testing (**Requirement 1.3**).

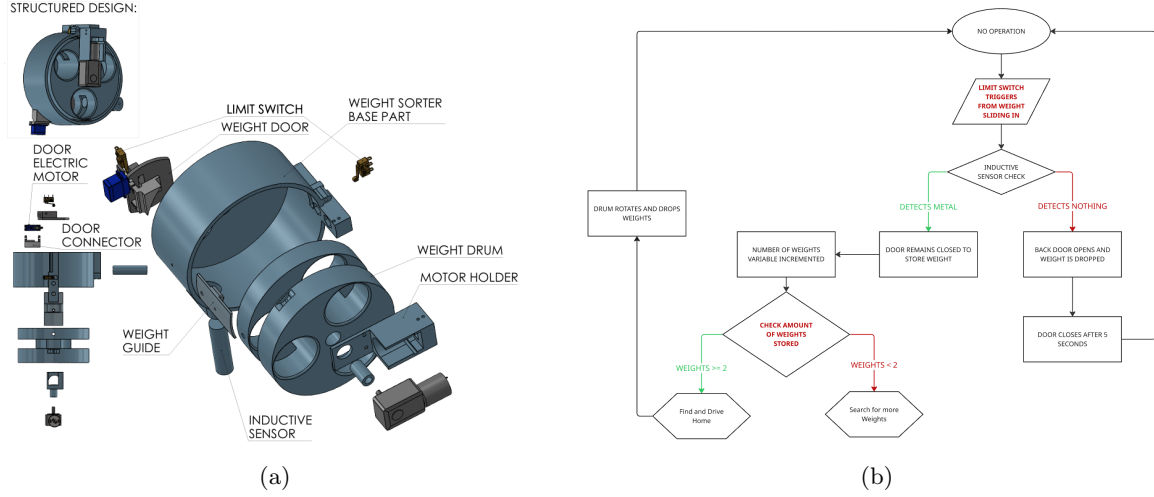


Figure 8: General details on weight storage system (a) and general logic tree for programming the storage mechanism (b)

2.5 Software

The navigation system integrates data from the Inertial Measurement Unit (IMU) and Time-of-Flight (TOF) distance sensors to generate a spatial representation of the arena. This representation is maintained as an integer matrix, enabling the identification of both static obstacles and free space. The arena, measuring 3m x 2m, is discretized into a 30 x 20 matrix, with each element corresponding to a 0.1m x 0.1m cell. At system initialization, all cells are initialized to a value of zero, indicating their current status as unknown. Robot positioning is estimated using a combination of wheel encoder data and IMU gyroscope readings. Encoder measurements of wheel rotation, together with the wheel radius, are used to determine translational displacement. This displacement is then integrated with gyroscopic heading data to estimate the robot's trajectory. IMU-derived position estimates are compared with encoder-based calculations to identify and compensate for discrepancies.

Once the robot's position and rotation is determined, TOF sensor measurements are used in conjunction with the robot's gyroscope reading to identify the corresponding matrix cell under observation, this conversion is displayed in Figure 9.

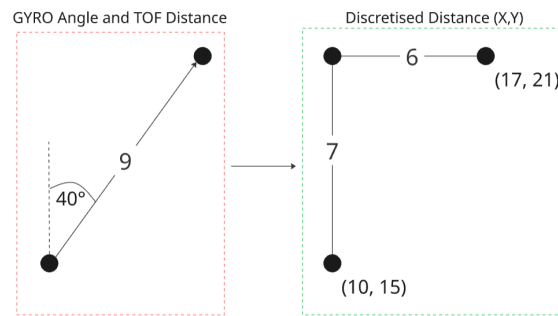


Figure 9: Conversion of IMU Gyro and TOF data into discretised X,Y coordinates

The measured distance is converted into Cartesian coordinates using trigonometric functions, accounting for the robot's current orientation. Bresenham's line algorithm is then applied between the robot's position and the detected cell to identify all intermediate coordinates. This process is shown in Figure 10 below, with the brown boxes demonstrating the matrix path that the robot would follow.

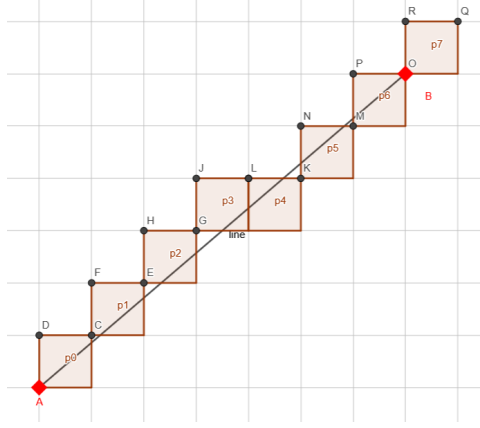


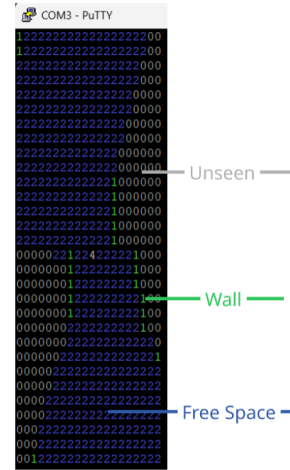
Figure 10: Bresenham Line Algorithm displaying intermediate boxes between points A and B.

Each intermediate cell is decremented by one, indicating a probable free space, while the detected cell is incremented by one, indicating a probable obstacle. Over multiple observations, consistent values become more confident as the data gets either more negative or more positive. This implementation is essential to reduce the influence of transient or moving objects.

The robot navigation matrix was tested by setting up a dummy arena inside of the Mechatronics Lab and having the robot rotate 360 degrees, centered in the middle of the arena. The arena is shown in Figure 11a, this is then printed to the terminal which is shown in Figure 11b.



(a)



(b)

Figure 11: Testing the robot navigation by turning the robot 360 degrees and comparing the original arena setup (a) to the matrix output (b)

The matrix displayed shows a large collection of the integer two displayed in blue color, these represent where the robot has seen free space, and should be able to travel to. Along with this the integer one displayed in green shows the walls placed within the arena, matching with the red walls shown in the real arena. In the center of the matrix a grey number 4 is displayed, representing where the robot currently is. The remaining cells are displayed as 0's meaning no data has been collected on these locations, this is due to the walls blocking the robots vision. A FSM was created for the predicted future control system, which is displayed below in Figure 12.

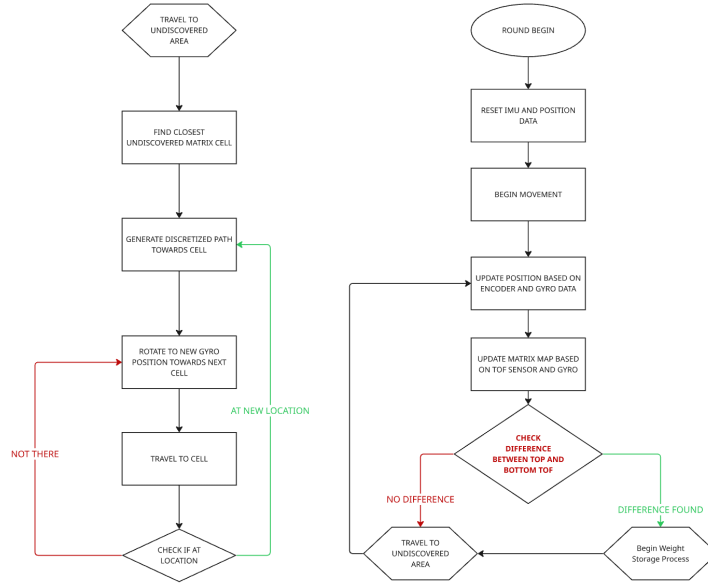


Figure 12: FSM of the current programming for movement around arena

The main loop is updating the current position of the robot using encoder values and the gyro rotation, this is then fed into the matrix map to determine the robots current position. If no weights are found, the robot needs to pick a new location to travel to.

2.6 Verification

The robot was tested and verified using the specifications defined in the previous report, these are displayed in Appendix A. The harvester, collection, and mechanical system were tested independently to determine worst-case scenarios, for which each test and its' result is listed in Appendix C. These tests would then be used to improve the individual features of the design to lessen any future unforeseen issues. A major verification issue found was the speed tests, where the alleged achievable speed was vastly over-estimated. The new estimate for success was now set to 2.9 km/h, for which this speed would be achievable with tuning of the track system.

For the other aspects of the design, the robot was found to have good mobility, being able to traverse in corners, up the ramp, and perform a nearly drift-less 360 degree turn. The design meets the area requirements, with the weight pickup and sorting very effective. The harvester, ramp and sorter work seamlessly together to allow for sub 5-second weight processing. The coding requirements are still under development, with the current progress made looking promising so far.

3 Fault Tree Analysis

An FTA has been used to understand and possibly minimise faults that could effect the system. The analysis has been included in Appendix D. Since the overall objective of the competition is to return to base with weights, this was used as the highest level failure event. From this, a top down approach was used to identify the root causes of major failure conditions. This separates failure modes into two separate scenarios; weight collection and return to base. These failures could be further broken down into three modes of failure; mechanical, electrical and software.

The impacts of the FTA were that there is considerable reliance on software operating without fault, as the majority of failures in collection or return to base came from software faults. This had the team make the decision to allow for majority effort to shift towards to software development, and there needs to be serious scrutiny on any code development. Another factor that was noted was quality in the wiring and soldering for all electrical components, which could lead to faults in all aspects of the design. This caused the team to evaluate all wiring connections and to check wiring quality. This also introduced the

long-term goal of efficient wire organisation, which is to lessen the probability and impact of a wire fault that could go undiagnosed during development or occur during the competition.

4 Remaining Tasks

4.1 Search Algorithm

The search algorithm is an on going component of the project, where the arena mapping has been finalised but it is not able to search within the space. Features to apply to the algorithm is incorporating weight detection and object avoidance sections into the code. These features would then be included into logic to allow the robot to explore areas that have not had weight detection conducted, to be able to navigate the arena through a SLAM-like system without any external assistance. This is important to allow for the robot to act independently and to not have to introduce any external features like a home beacon which would increase the variables that would have to be managed.

4.2 Object Detection

Object detection is currently being developed, with two options of IR brightest point or TOF triangulation system being under consideration. Either of these systems is extremely important to help with the overarching search and navigation algorithm, to allow for objects to be detected and then collected by the robot. There will need to be co-ordination between the object detection and navigation sensors, to reduce the possibility of either system overriding each other which could result in uncollected weights or logic errors that could stop the robot functioning as displayed in the FTA.

4.3 Sensor Filtering

Currently, the TOF algorithm is updated every 50 milliseconds, no filtering or buffer is applied and the data is immediately transmitted to the Bresenham algorithm mentioned earlier. Due to this, the noise from the TOF and IMU sensors vary marginally within cells, resulting in incomplete data or high variable information. These same issues will likely be experienced with the TOF sensors used in the weight detection. The current intention is to implement a moving average filter into the code. However, as the navigation mapping will use a large amount of CPU load, an analogue low pass filter using a RC circuit may be used instead. These improvements are to ensure that the best quality information is then being fed to the software, not only for the algorithm but also to make debugging issues with navigation easier into the future, which was highlighted by the FTA.

4.4 Navigation Sensors

Currently the current navigation system uses one TOF sensor at the front to detect objects. This leads to slower sensing as the updated matrix only has one piece of data per loop to update. The intention is to implement two more TOF sensors to the front of the robot. These will be used in the same way as the single TOF sensors, reducing blind spots on the robot, and allowing the arena to be mapped faster.

4.5 Sensor Mounts

At this stage, the single TOF sensor is held onto the robot using tape, and the other distance sensors have not been attached. While testing needs to be completed for optimal placement, the chassis will need to be modified to connect sensor mounts. This will increase the reliability of the sensors as they will be held in a consistent place, and risk of damage to the sensors will be reduced.

4.6 Collection System

Currently the harvester collection system is capable of collecting weights in any orientation assuming that the weights are centered. However, forward facing weights that enter near the sides of the harvester get caught against the ramp, jamming the harvester entirely. On planned improvement will be to reduce the risk in the code, by ensuring the robot is lined up with weights before attempting collection. the mechanical design will also be iterated to increase the usable area of the harvester.

4.7 Sorting System

At this stage the mechanical assembly and electronics have been completed on the robot. This system is yet to be integrated into the code and tested with the harvester. While smooth movement of the weights is expected, there are multiple potential areas where weights or the revolver may get jammed. Further testing is required to be confident in the reliability of the system.

4.8 Movement Detection

Currently, the robot detects the arena that surrounds it through the TOF sensors and the IMU gyro positioning. Further implementation is required to create a system where the encoder values and IMU position are used in collaboration with each other to enable confident position detection. This position will be used inside the navigation system to analyse where the robot has been in order to create an optimized path system to improve the overall search algorithm.

4.9 Track Friction

With the current design of the tracks, rubbing occurs between the tracks and the mount of the collection system. While this currently works, the added friction is causing unnecessary wear, and will be reducing the overall speed of the robots movement. Neither of these components can be moved significantly with the current design. To reduce the rubbing between these, a silicon layer will be applied to reduce friction. This should allow the motors to spin faster and wear down the tracks at a slower pace.

5 Conclusion

The initial build of the mechanical design has been fully completed. This includes the chassis, movement, collection mechanism, sorting mechanism, and weight storage. The software navigation is capable of generating a spatial representation of the arena based on TOF, wheel encoder, and IMU data. Most mechanical and electrical systems have encountered minimal changes from the initial concept design. The sorting system and weight collection is an exception to this as it has taken a significant design direction due to space restrictions. Potential failure modes and concerns have been identified using a qualitative fault tree analysis. These potential risks have been minimised through the direction and discussion of the remaining tasks. At this stage the mechanical systems require ongoing testing and tweaks to improve the reliability and consistency of the robot's performance. The electrical and software systems still require major aspects to be started to achieve full performance. This includes object detection, an integrated search algorithm, and integration of every module to function at once. The software will have to be carefully monitored as to ensure to reduce any possible bugs or faults that could occur during the match. Overall, the project is on target to be completed with full functionality prior to the competition.

A Robot Updated Specifications

1. Performance Requirements

- 1.1 The robot should be able to pick up weights up to 1 kg (without need of initial discrimination)
Rationale: There is not enough available time in the match nor reliable enough design to try to discriminate the weight's mass and before pick up.
- 1.2 The robot should be able to detect a weight within 250mm of itself (without need of initial discrimination)
Rationale: The robot needs to have some guidance to weights to make it more effective at weight pickup.
- 1.3 The design must be able to determine what type of weight it has picked up in a maximum 5 second time frame
Rationale: If there are multiple weights that the robot encounters, the weight processing must be as efficient as possible.
- 1.4 The robot must be able to capture the snatch, even if that does sacrifice performance.
Rationale: The snatch is a valuable asset to capture during the competition, and its size makes it quite difficult to properly process without increased dimensional constraints.
- 1.5 The robot must be able to climb a 30 °incline and a maximum height of 100mm
Rationale: Previous competitors had large issues with ramps leading to large failures

2. Functional Requirements

- 2.1 The robot should be able to turn around in any corners without getting stuck
- 2.2 The robot shall be capable of turning 360 degrees while remaining in place
Rationale: These features allow the robot to be able to reorientate itself at any location in the arena
- 2.3 The robots' track system must be able to move the robot at walking pace (2.9km/h) when fully loaded.
Rationale: The time pressure of the match is a major factor, the faster the robot can traverse the course, the faster it can collect weights when considering competition.
- 2.4 Weights must be able to be picked up no matter orientation into the system
Rationale: During a round weights are knocked over and moved, this leads to a variety of orientations that would need to be picked up
- 2.5 The robot must keep track of the amount of weights on board the robot
Rationale: Holding more than 2 weights at the end of the round will lead to point deductions.

3. Operational Requirements

- 3.1 All components of the robot, mechanical or electrical, must be able to function for 10 minutes or more.
Rationale: The match only lasts a fraction of the time, however stress testing the robot with longer times and introducing stressful situations on the system may present deficiencies that can be fixed before the proper competition.

4. Non-functional Requirements

- 4.1 The robot's components must be easily removed and replaced to allow for modularity
Rationale: The time between rounds is short, if modification needs to be made it needs to be efficient and quick

5. Constraints

- 5.1 The robot must fit in a 400x400mm area.
Rationale: The tightest corners of the design are within that 400mm size, so a robot that can traverse in tight areas like that will be able to go across majority of the course fine.
- 5.2 All parts fabricated should be able to be fabricated at university with no external help or specific constraints.
Rationale: If failures occur during the competition, parts would need to be fabricated within the university.

B Robot Progress Evaluation Results Table

Table 1: Evaluation table for robot qualities testing for mechanical system

Robot Verification Analysis		
Specification	Importance	Rating
1.1 - Pick up heavy weights	10	Completed
1.2 - Detect close weights	7	Not completed
1.3 - Determine type of weight	6	In Progress
1.4 - Capture the snitch	4	Completed
1.5 - Climb inclines effectively	5	Completed
2.1 - Turn in corners	8	Completed
2.2 - Perform 360 without moving	6	Completed
2.3 - Move at fast speed	6	Failed
2.4 - Pick up any orientation	10	Completed
2.5 - Keep track of weight count	9	In Progress
3.1 - Components last at least 10 minutes	5	In Progress
4.1 - Modular robot assembly	8	In Progress
5.1 - Fit within a 400x400mm area	10	Completed
5.2 - Parts fabricated at the university	7	Completed

C Design Requirement Testing and Verification

1.1 - Pick up Heavy Weights Weight collection was tested for each type of weights. The robot was set to drive in a straight line with the harvester spinning to simulate competition collection. The sorting system was removed to isolate the collector system only. Weight collection was tested both with the weights going through the middle of the harvester and the side of the harvester. All main orientations of weights were tested. All weights could be picked up in the sideways orientation through the middle and side of the harvester. All upright and forward facing weights could be picked up through the middle of the harvester, but would get caught on the lip of the ramp at the edges of the harvester.

1.2 Detect Close Weights Not tested - Weight detection not completed at this stage.

1.3 - Determine Type of Weight The sorting system weight type detection has been testing in isolation using the proximity sensor. This was achieved by sliding in weights while holding the sorting system and checking for a counter to be updated while connected to a computer. This yielded stable results with correct calibration and mechanical tuning of the sorter. This system has not been tested while installed on the robot with the correct orientation. Further testing required to consider design requirement met.

1.4 - Capture the Snitch Snitch collection was tested using a reduced arena size. Two separate situational test were completed to simulate the true arena environment. For the first test, the robot was set stationary with the harvester spinning. The snitch was rolled directly into the harvester. For

the second test, the snitch was left stationary and the robot drove directly toward the snitch with the harvester spinning. Both tests resulted in the snitch being picked up.

1.5 - Climb Inclines Effectively The robot was tested with the tracks set to drive directly forward with the collection system, navigation, and sorting system not operational. The robot was sent over both the starting area bump, and up the arena ramp. In both scenarios the robot was capable of driving over without getting stuck or tipping.

2.1 - Turn in Corners To test, the robot was set to drive on a preplanned movement. This included driving into a 400x400 corner made using arena wall obstacles. The robot then had to spin 180 degrees and drive out without getting stuck. After multiple trials the robot was capable of completing the preplanned path without getting stuck.

2.2 - Perform 360 Without Moving For testing this situation, the robot was placed in parts of the arena with the following constraints to understand the effect of any possible drift of the robot whilst turning. This also was to demonstrate the effect that the robot might have on the arena itself

- The robot was placed center in the arena with no obstacles in its path.
- The robot was placed close to the black walls of the arena with spacing of 20mm from edge of the robot to the wall.
- The robot was placed next to one pillar that was spaced 20mm from the edge of the robot.
- The robot was placed between two movable walls with 20mm of space on either side.
- The robot was placed at the angled stages of the ramp to understand if it could have minimal slippage under the effect of its own weight.
- The robot was placed at the level stage of the ramp to see if it would fall off the ramp, and consequentially tip over afterwards.

Each of these situations were tested 3 times, and it was found that the robot was able to complete majority of these challenges. However for the tipping aspect there was one occasion where the robot did tip over and the robot did not fall off the ramp track first. However the risk of this occasion happening again was deemed minimal as having the turn on the ramp is not an advised action.

2.3 - Move at Fast Speed The robot was given the distance of the arena to travel from side to side, with recording the speed of the robot through calculating the time it took.

2.4 - Pick up in Any Orientation This test was covered during test 1.1 - Pick up Heavy Weights.

2.5 - Keep Track of Weight Count The general weight counter was tested in 1.3 - Determine Type of Weight. This has not been tested as fully integrated code.

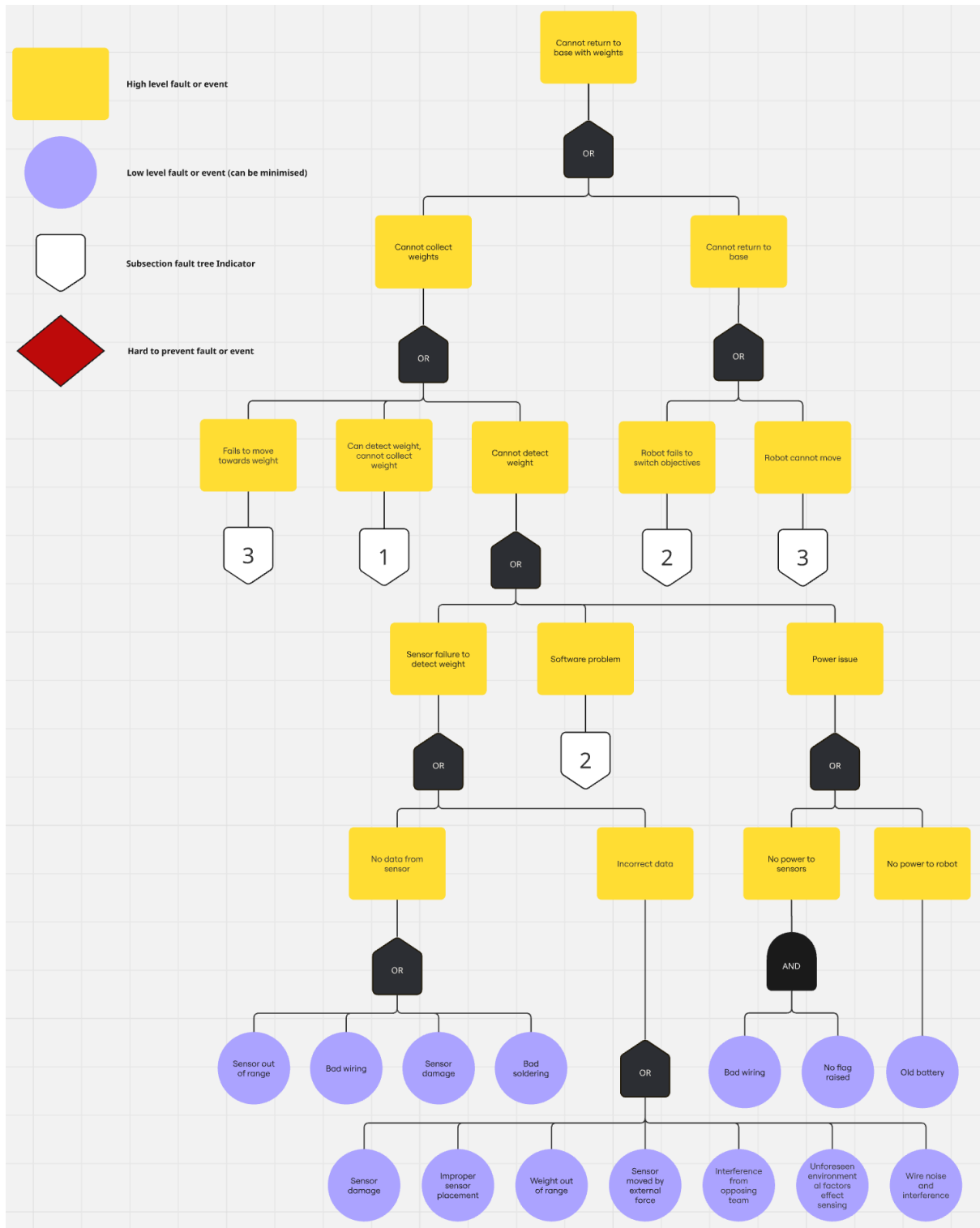
3.1 - Components Last at Least 10 Minutes The robot was run with all motor for 10 minutes. The robot is capable of doing this without draining the battery. Not all components and code are used on the robot yet; thus, further testing will be required in the future.

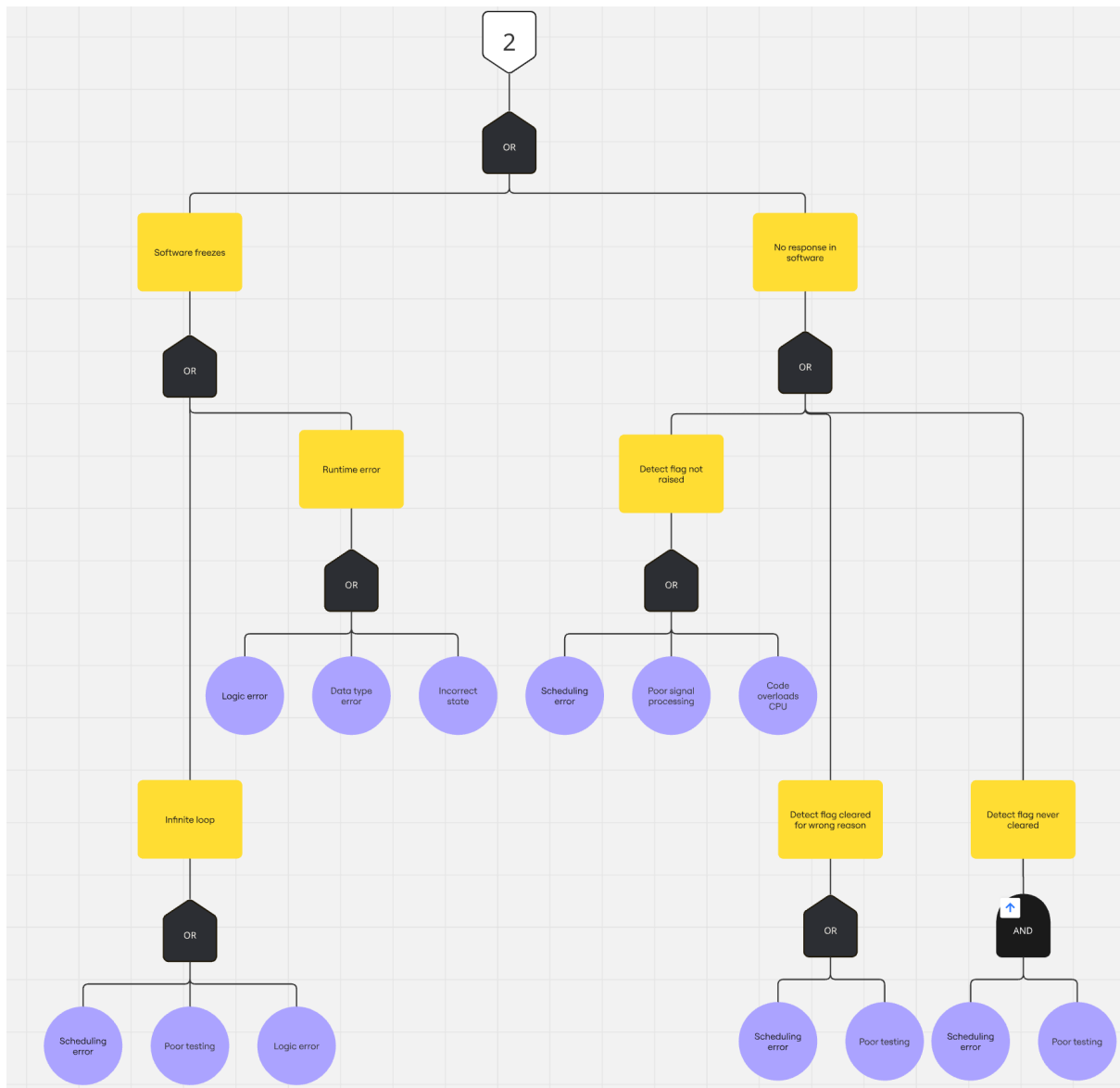
4.1 - Modular Robot Assembly Each section of the robot has been tested for modularity. The sorter, sensors, and collector passed the modularity test, as they could be removed without taking apart other aspects of the robot. The tracks were unable to pass the modularity test as they are heavily integrated into the chassis of the robot.

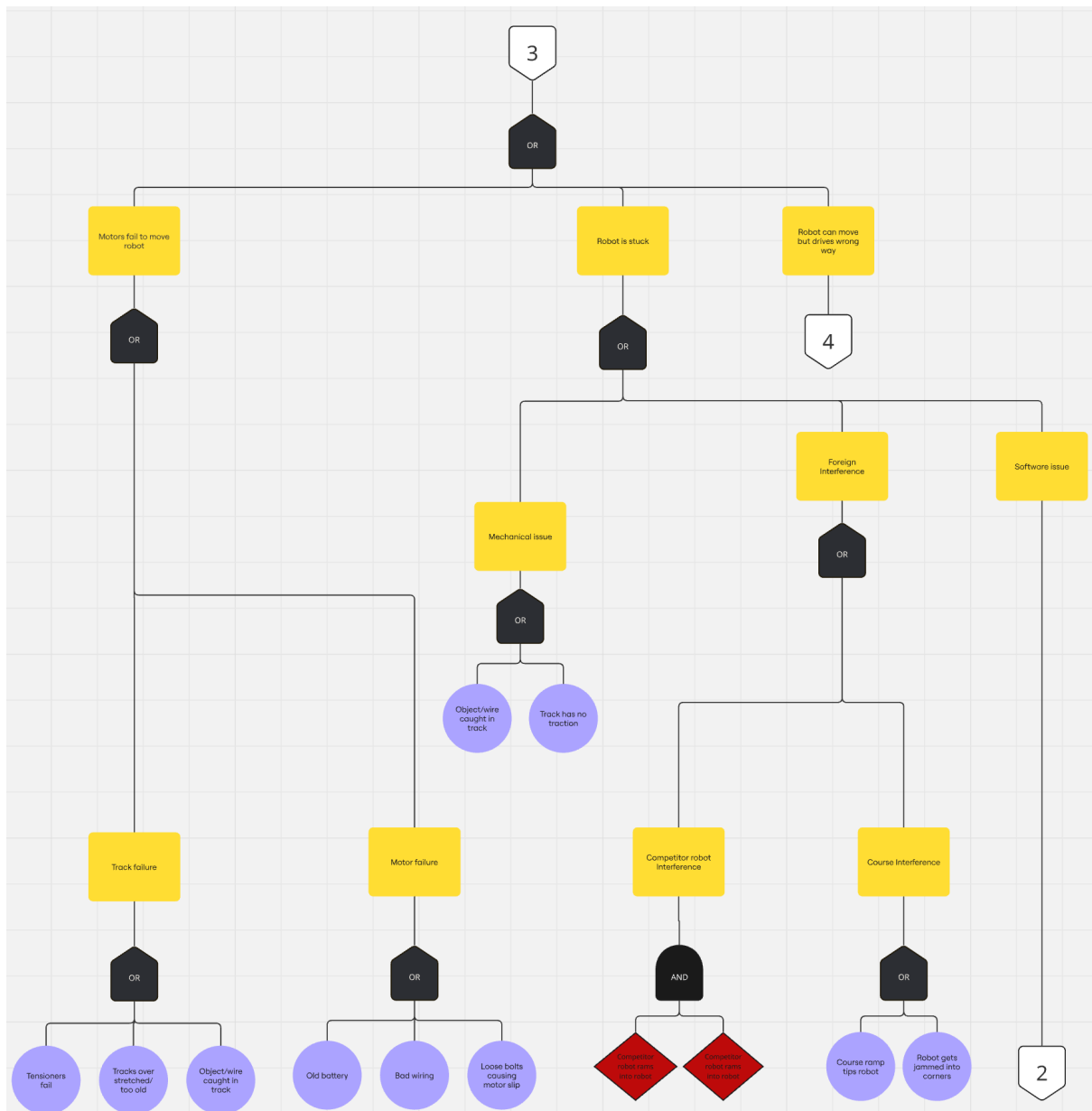
5.1 - Fit Within a 400x400 Area This test was covered during 2.1 - Turn in corners.

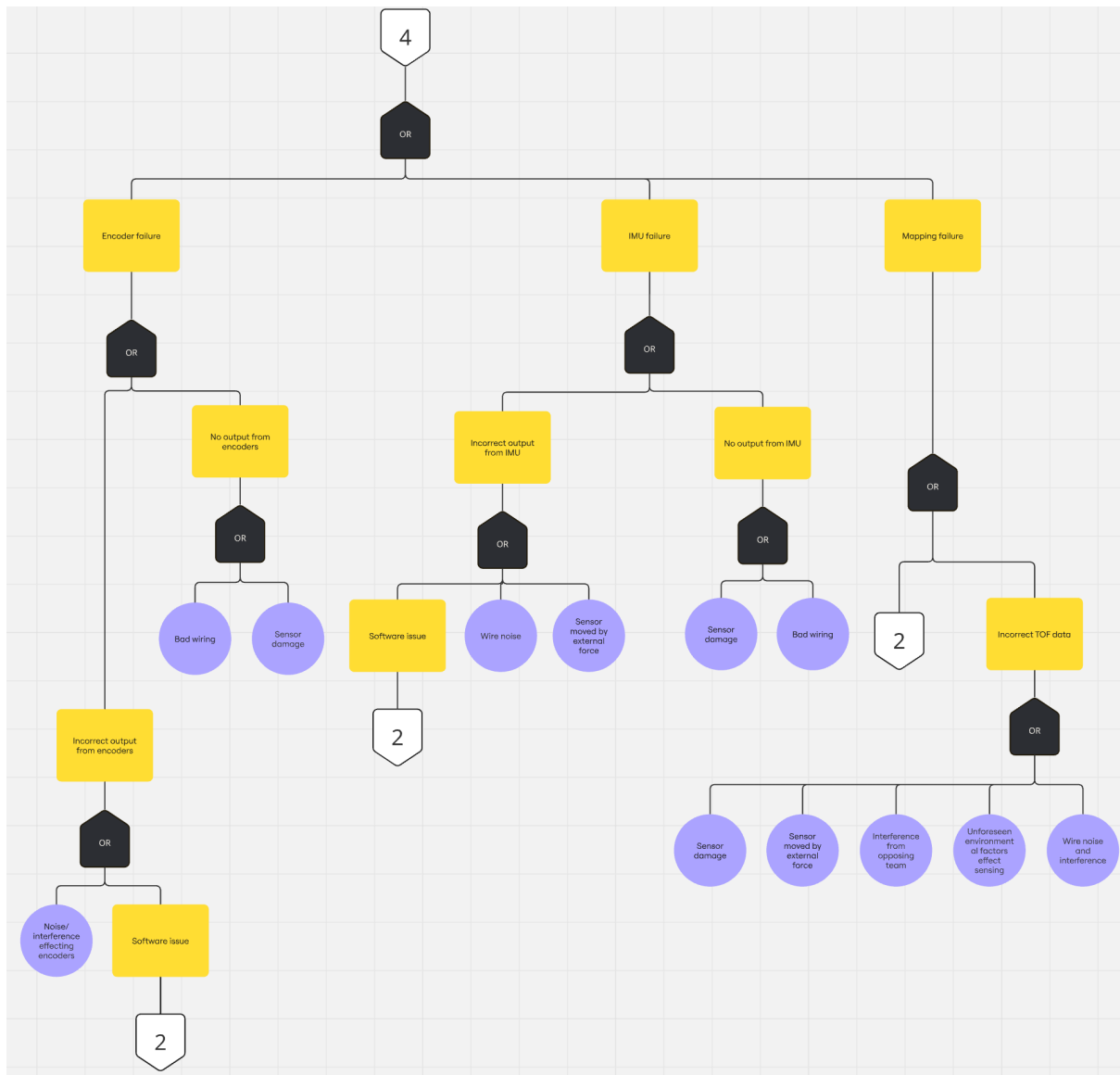
5.2 - Only Fabricated Parts at the University All parts thus far have been fabricated at the university of provided as part of the Robocup components kit.

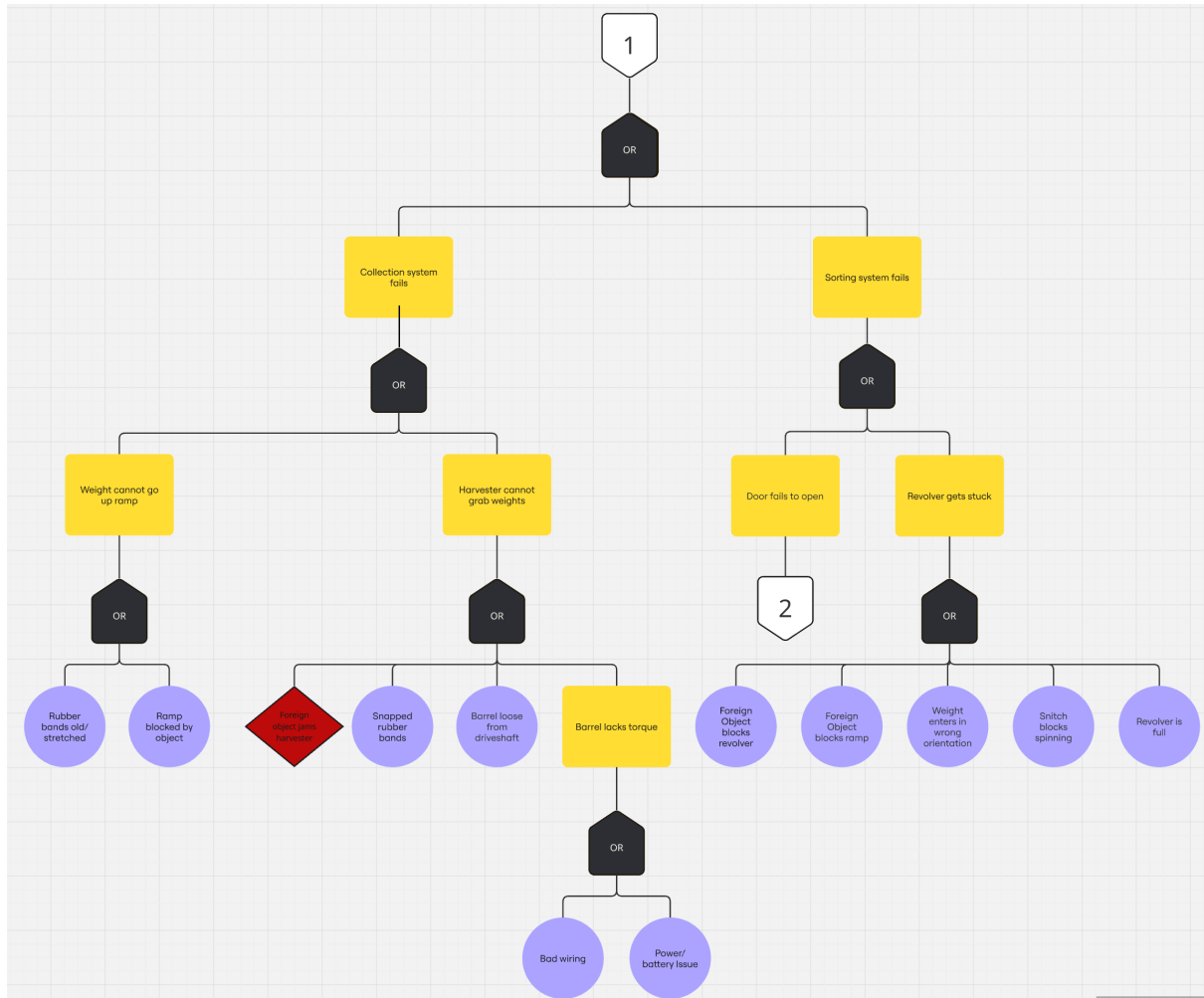
D FTA Table





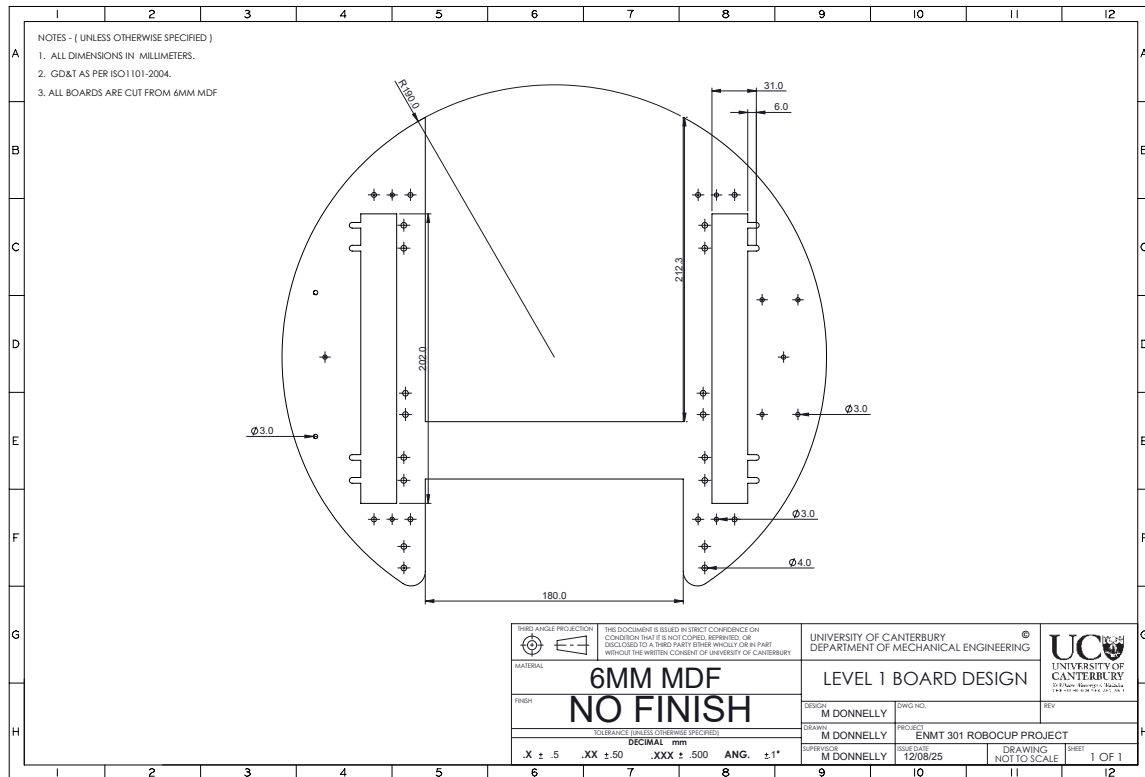




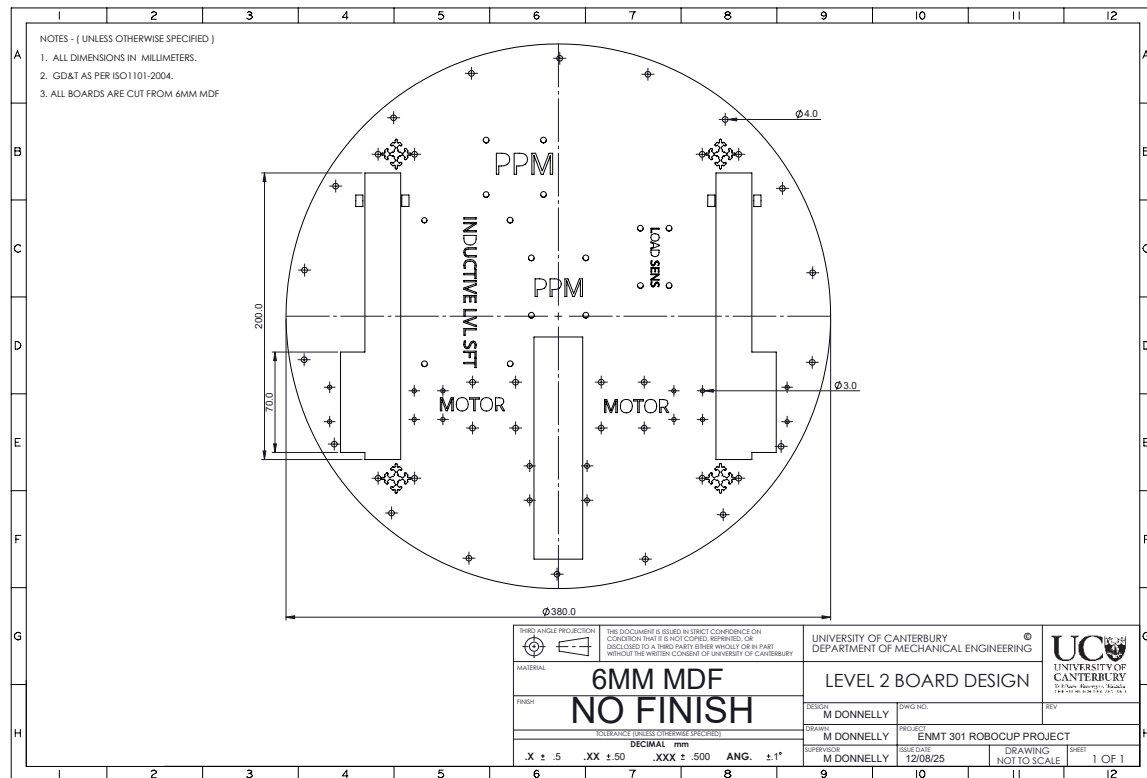


E Engineering drawings

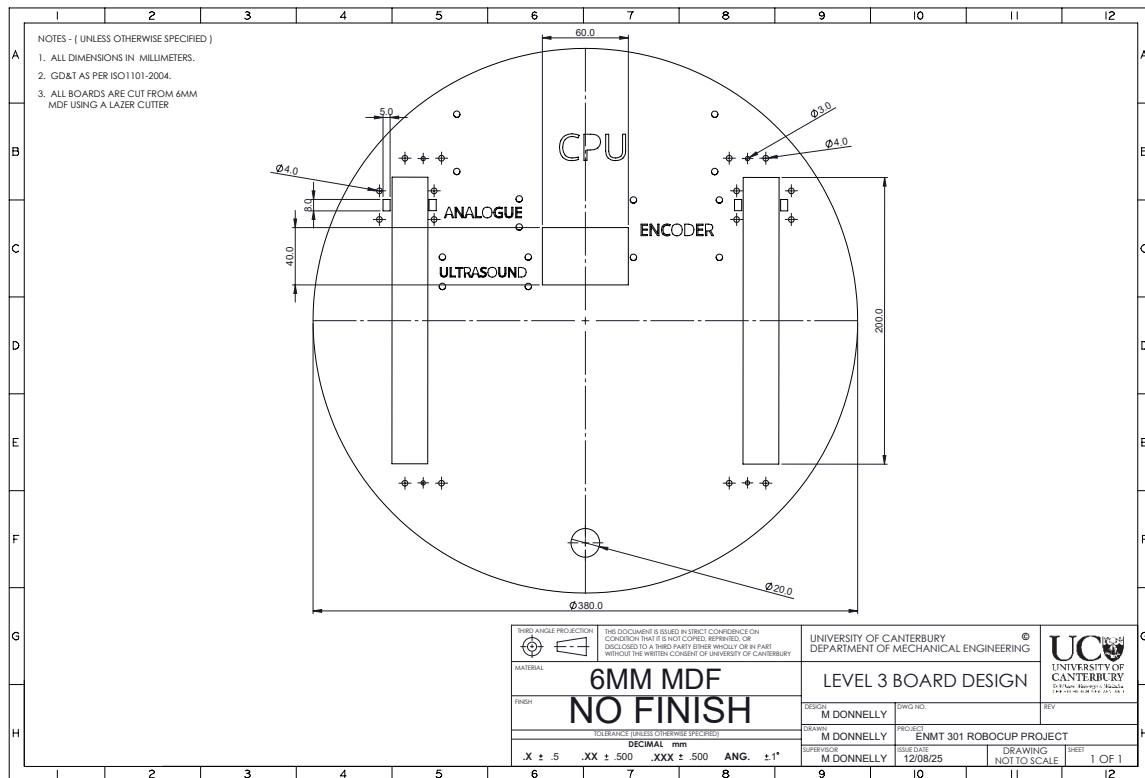
E.1 Level 3 Design



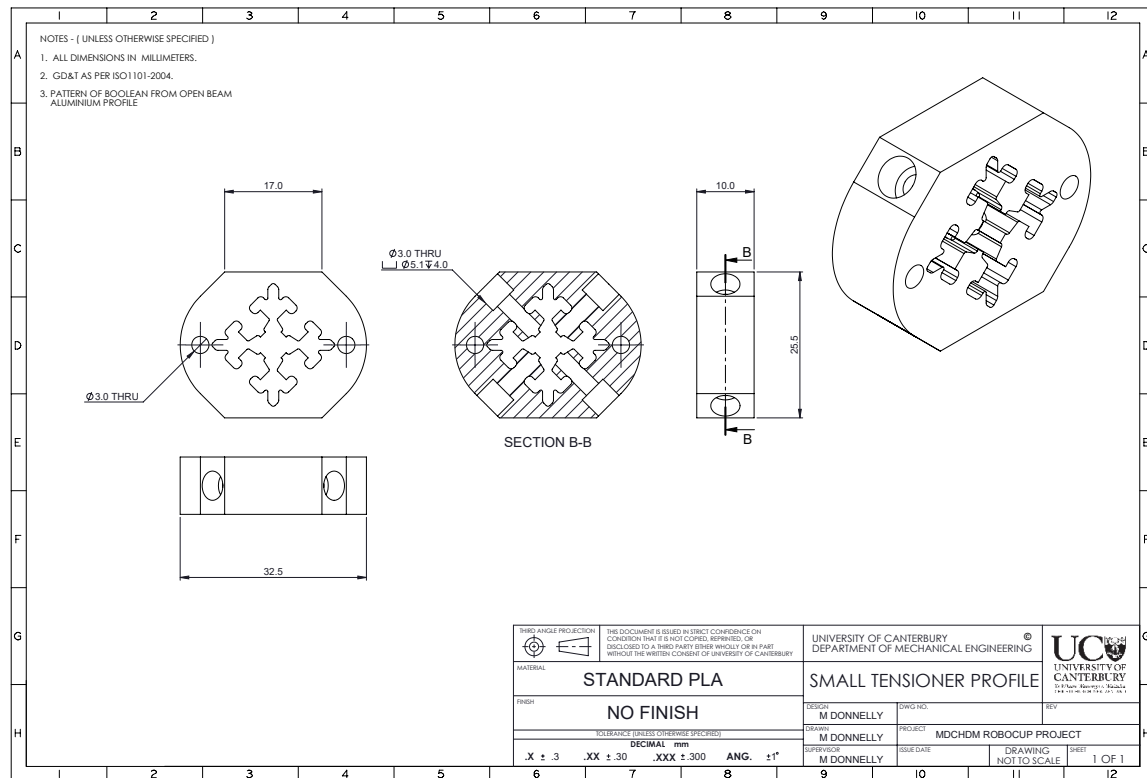
E.2 Level 2 Design



E.3 Level 1 Design



E.4 Tensioner Profile Design



E.5 Harvester Wheel design

